

Data Dictionary

Full Dataset (Including Children)

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December 2025

Table 1: Complete data dictionary for the Swiss National CI Database (Full dataset including children, N=771). Variables sorted by category.

Variable	Category	Type	N (%)	Description
Age_at_OP	Demographics	Num	771 (100%)	Age at CI surgery (years)
GebDatum	Demographics	Date	771 (100%)	Date of birth
Geschlecht	Demographics	Cat	771 (100%)	Sex (M/F)
Nation	Demographics	Cat	95 (12%)	Nationality
Nation_Mu	Demographics	Cat	583 (76%)	Mother nationality
Nation_Va	Demographics	Cat	572 (74%)	Father nationality
PLZ	Demographics	Num	771 (100%)	Postal code
PatNr	Demographics	Cat	771 (100%)	Patient ID
verstorben	Demographics	Cat	771 (100%)	Deceased
C12_post	Speech	Num	500 (65%)	Post-op consonant test (%)
C12_pre	Speech	Num	304 (39%)	Pre-op consonant test (%)
Date_PTA_preOP_CI	Speech	Date	704 (91%)	
Date_PTA_preOP_Co	Speech	Date	705 (91%)	
FM_post	Speech	Num	495 (64%)	Post-op Freiburg monosyllables (%)
FM_pre	Speech	Num	328 (43%)	Pre-op Freiburg monosyllables (%)

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Variable	Category	Type	N (%)	Description
Mono1_post	Speech	Num	464 (60%)	Post-op monosyllable word score (%)
Mono1_pre	Speech	Num	307 (40%)	Pre-op monosyllable word score (%)
Mono2_post	Speech	Num	464 (60%)	Post-op monosyllable score 2 (%)
Mono2_pre	Speech	Num	307 (40%)	Pre-op monosyllable score 2 (%)
Mono_bi1_post	Speech	Num	17 (2%)	Post-op number test (%)
Num_post	Speech	Num	488 (63%)	
Num_pre	Speech	Num	318 (41%)	Pre-op number test (%)
Remark_FF_pre_CI	Speech	Cat	399 (52%)	
Remark_PTA_pre_CI	Speech	Cat	482 (63%)	
Remark_PTA_pre_Co	Speech	Cat	478 (62%)	
V08_post	Speech	Num	510 (66%)	Post-op OLSA sentence test (%)
V08_pre	Speech	Num	314 (41%)	Pre-op OLSA sentence test (%)
prBONE_1000_CI	Audiometry	Num	570 (74%)	Bone conduction 1000Hz CI (dB)
prBONE_1000_Co	Audiometry	Num	572 (74%)	Bone conduction 1000Hz contra (dB)
prBONE_2000_CI	Audiometry	Num	572 (74%)	Bone conduction 2000Hz CI (dB)
prBONE_2000_Co	Audiometry	Num	565 (73%)	Bone conduction 2000Hz contra (dB)
prBONE_250_CI	Audiometry	Num	551 (71%)	Bone conduction 250Hz CI (dB)

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Variable	Category	Type	N (%)	Description
prBONE_250_Co	Audiometry	Num	557 (72%)	Bone conduction 250Hz contra (dB)
prBONE_4000_CI	Audiometry	Num	574 (74%)	Bone conduction 4000Hz CI (dB)
prBONE_4000_Co	Audiometry	Num	568 (74%)	Bone conduction 4000Hz contra (dB)
prBONE_500_CI	Audiometry	Num	563 (73%)	Bone conduction 500Hz CI (dB)
prBONE_500_Co	Audiometry	Num	567 (74%)	Bone conduction 500Hz contra (dB)
prFF_1000_CI	Audiometry	Num	342 (44%)	Free-field 1000Hz CI (dB)
prFF_1000_Co	Audiometry	Num	370 (48%)	Free-field 1000Hz contra (dB)
prFF_2000_CI	Audiometry	Num	340 (44%)	Free-field 2000Hz CI (dB)
prFF_2000_Co	Audiometry	Num	359 (47%)	Free-field 2000Hz contra (dB)
prFF_250_CI	Audiometry	Num	329 (43%)	Free-field 250Hz CI (dB)
prFF_250_Co	Audiometry	Num	355 (46%)	Free-field 250Hz contra (dB)
prFF_4000_CI	Audiometry	Num	331 (43%)	Free-field 4000Hz CI (dB)
prFF_4000_Co	Audiometry	Num	354 (46%)	Free-field 4000Hz contra (dB)
prFF_500_CI	Audiometry	Num	340 (44%)	Free-field 500Hz CI (dB)
prFF_500_Co	Audiometry	Num	366 (47%)	Free-field 500Hz contra (dB)
prFF_6000_CI	Audiometry	Num	294 (38%)	Free-field 6000Hz CI (dB)
prFF_6000_Co	Audiometry	Num	322 (42%)	Free-field 6000Hz contra (dB)

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Variable	Category	Type	N (%)	Description
prFF_8000_CI	Audiometry	Num	53 (7%)	
prFF_8000_Co	Audiometry	Num	54 (7%)	
prPTA_1000_CI	Audiometry	Num	704 (91%)	PTA 1000Hz CI ear (dB)
prPTA_1000_Co	Audiometry	Num	705 (91%)	PTA 1000Hz contralateral (dB)
prPTA_125_CI	Audiometry	Num	618 (80%)	PTA 125Hz CI ear (dB)
prPTA_125_Co	Audiometry	Num	629 (82%)	PTA 125Hz contralateral (dB)
prPTA_2000_CI	Audiometry	Num	699 (91%)	PTA 2000Hz CI ear (dB)
prPTA_2000_Co	Audiometry	Num	695 (90%)	PTA 2000Hz contralateral (dB)
prPTA_250_CI	Audiometry	Num	681 (88%)	PTA 250Hz CI ear (dB)
prPTA_250_Co	Audiometry	Num	675 (88%)	PTA 250Hz contralateral (dB)
prPTA_4000_CI	Audiometry	Num	695 (90%)	PTA 4000Hz CI ear (dB)
prPTA_4000_Co	Audiometry	Num	685 (89%)	PTA 4000Hz contralateral (dB)
prPTA_500_CI	Audiometry	Num	700 (91%)	PTA 500Hz CI ear (dB)
prPTA_500_Co	Audiometry	Num	697 (90%)	PTA 500Hz contralateral (dB)
prPTA_6000_CI	Audiometry	Num	577 (75%)	PTA 6000Hz CI ear (dB)
prPTA_6000_Co	Audiometry	Num	586 (76%)	PTA 6000Hz contralateral (dB)
prPTA_8000_CI	Audiometry	Num	550 (71%)	PTA 8000Hz CI ear (dB)

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Variable	Category	Type	N (%)	Description
prPTA_8000_Co	Audiometry	Num	562 (73%)	PTA 8000Hz contralateral (dB)
COM_ARTICULATION	Communication	Cat	721 (94%)	Articulation quality
COM_FATHER_DEAF	Communication	Cat	672 (87%)	Father deaf
COM_FATHER_HACI_USE	Communication	Cat	234 (30%)	Father sign language use
COM_Gebaerden	Communication	Cat	718 (93%)	Sign language use
COM_LANG_SCHOOL	Communication	Cat	65 (8%)	Signed German use
COM_LBG	Communication	Cat	717 (93%)	
COM_LIP_READING	Communication	Cat	431 (56%)	Lip reading ability
COM_Lautspra	Communication	Cat	759 (98%)	Spoken language use
COM_MAIN_LANGUAGE	Communication	Cat	762 (99%)	Primary language
COM_MOTHER_DEAF	Communication	Cat	667 (87%)	Mother deaf
COM_MOTHER_HACI_USE	Communication	Cat	229 (30%)	Mother sign language use
COM_MULTILANG_HOME	Communication	Num	756 (98%)	Multilingual home
COM_PHONE_USE	Communication	Cat	554 (72%)	Phone use ability
COM_Profession	Communication	Cat	327 (42%)	Secondary language
COM_SECONDARY_LANGUAGE	Communication	Cat	267 (35%)	
COM_Total	Communication	Cat	709 (92%)	Total communication

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Variable	Category	Type	N (%)	Description
EVA_DATE_CI_DIAG	Evaluation	Date	414 (54%)	Date of CI candidacy
EVA_DATE_CI_DIAG_L	Evaluation	Date	314 (41%)	
EVA_DATE_HAID_1	Evaluation	Date	524 (68%)	Date of first hearing aid
EVA_DATE_HAID_1_L	Evaluation	Date	512 (66%)	
EVA_DATE_HL_DIAG	Evaluation	Date	510 (66%)	Date of HL diagnosis
EVA_DATE_HL_DIAG_L	Evaluation	Date	477 (62%)	Date of HL diagnosis (left)
EVA_DATE_TOTAL_DEAF	Evaluation	Date	53 (7%)	
EVA_DATE_TOTAL_DEAF_L	Evaluation	Date	16 (2%)	
EVA_DEAF_ONSET	Evaluation	Cat	618 (80%)	Onset of deafness
EVA_DEAF_ONSET_L	Evaluation	Cat	468 (61%)	Onset of deafness (left)
EVA_ETIOLOGY	Evaluation	Cat	771 (100%)	Etiology of hearing loss
EVA_ETIOLOGY_L	Evaluation	Cat	709 (92%)	Etiology (left ear)
EVA_ETIOLOGY_TEXT	Evaluation	Cat	185 (24%)	
EVA_Eclampsia	Evaluation	Cat	635 (82%)	Eclampsia
EVA_HL_PROGREDIENTE	Evaluation	Cat	626 (81%)	Progressive hearing loss
EVA_HL_PROGREDIENTE_L	Evaluation	Cat	609 (79%)	
EVA_Intraut_infect	Evaluation	Cat	637 (83%)	Intrauterine infection
EVA_KNOWS_CIPAT	Evaluation	Cat	696 (90%)	Knows other CI patients

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Variable	Category	Type	N (%)	Description
EVA_Pregn_normal	Evaluation	Cat	209 (27%)	Normal pregnancy
EVA_SUBJ_CI_DIAG	Evaluation	Cat	635 (82%)	Subjective CI diagnosis
EVA_SUBJ_CI_DIAG_L	Evaluation	Cat	623 (81%)	
EVA_SUBJ_HAID_1	Evaluation	Cat	645 (84%)	Subjective hearing aid use
EVA_SUBJ_HAID_1_L	Evaluation	Cat	626 (81%)	
EVA_SUBJ_HL_DIAG	Evaluation	Cat	636 (82%)	Subjective HL diagnosis
EVA_SUBJ_HL_DIAG_L	Evaluation	Cat	617 (80%)	
EVA_born_at	Evaluation	Num	42 (5%)	
EVA_other	Evaluation	Cat	637 (83%)	
SES_AHV	Socioeconomic	Cat	650 (84%)	AHV (old age insurance)
SES_Alimente	Socioeconomic	Cat	648 (84%)	
SES_Andere	Socioeconomic	Cat	641 (83%)	
SES_Arbeitslos	Socioeconomic	Cat	646 (84%)	
SES_CARE_GIVER	Socioeconomic	Cat	212 (27%)	
SES_EDUCATION_LEVELS	Socioeconomic	Cat	489 (63%)	Education level
SES_EDUC_LEVEL_FATHER	Socioeconomic	Cat	200 (26%)	
SES_EDUC_LEVEL_MOTHER	Socioeconomic	Cat	190 (25%)	

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Variable	Category	Type	N (%)	Description
SES_EDUC_SITUATION	Socioeconomic	Cat	166 (22%)	
SES_EL	Socioeconomic	Cat	646 (84%)	
SES_EMPLOYMENT_TYPE	Socioeconomic	Cat	251 (33%)	
SES_HEALTH_INSURANCE	Socioeconomic	Cat	734 (95%)	Health insurance
SES_Hilfl	Socioeconomic	Cat	647 (84%)	
SES_IV	Socioeconomic	Cat	648 (84%)	IV (disability insurance)
SES_LIVING_SITUATIONS	Socioeconomic	Cat	426 (55%)	Living situation
SES_MARITIAL_STATE	Socioeconomic	Cat	651 (84%)	Marital status
SES_ORPHAN	Socioeconomic	Num	0 (0%)	
SES_ORPHAN_ADOPTED	Socioeconomic	Cat	116 (15%)	
SES_PROFESSION_LEARNED	Socioeconomic	Cat	456 (59%)	Learned profession
SES_PROF_LEARNED_FATHER	Socioeconomic	Cat	186 (24%)	
SES_PROF_LEARNED_MOTHER	Socioeconomic	Cat	174 (23%)	
SES_PROF_PENSUM	Socioeconomic	Num	240 (31%)	Work percentage
SES_PROF_POSITION	Socioeconomic	Cat	196 (25%)	
SES_REASON_NOT_WORKING	Socioeconomic	Cat	284 (37%)	
SES_RENT_PARTIAL	Socioeconomic	Num	107 (14%)	

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Variable	Category	Type	N (%)	Description
SES_RES_PERMIT_SINCE	Socioeconomic	Date	465 (60%)	Residence permit status
SES_RES_PERMIT_STATUS	Socioeconomic	Cat	656 (85%)	
SES_SCHOOL_TYPE	Socioeconomic	Cat	43 (6%)	
SES_SCHOOL_YEAR	Socioeconomic	Cat	37 (5%)	Currently working
SES_SUVA	Socioeconomic	Cat	646 (84%)	
SES_Soz	Socioeconomic	Cat	646 (84%)	
SES_Stip	Socioeconomic	Cat	647 (84%)	
SES_Tag	Socioeconomic	Cat	646 (84%)	
SES_Ueberbr	Socioeconomic	Cat	645 (84%)	
SES_WORKING	Socioeconomic	Cat	605 (78%)	
SES_Witwe	Socioeconomic	Cat	646 (84%)	
CT_Brain_path	CT	Cat	486 (63%)	Brain pathology on CT
CT_Carotid_a_abnorm	CT	Cat	492 (64%)	Carotid artery abnormal
CT_Cochl_malform	CT	Cat	498 (65%)	Cochlear malformation
CT_Cochlear_otoscl	CT	Cat	490 (64%)	Cochlear otosclerosis
CT_Laby_ossi	CT	Cat	488 (63%)	Labyrinthine ossification
CT_MRI_Other	CT	Cat	420 (54%)	Other CT/MRI findings

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Variable	Category	Type	N (%)	Description
CT_Mastoid_middle_ear_abnorm	CT	Cat	494 (64%)	Mastoid/middle ear abnormality
CT_Pathway_VII_abnorm	CT	Cat	492 (64%)	Facial nerve pathway abnormal
CT_Sigmoid_sin_abnorm	CT	Cat	492 (64%)	Sigmoid sinus abnormal
CT_Trauma	CT	Cat	488 (63%)	Trauma on CT
CT_done	CT	Cat	639 (83%)	CT performed
MRI_Brain_path	MRI	Cat	605 (78%)	Brain pathology on MRI
MRI_Cochl_Nerve_absence	MRI	Cat	602 (78%)	Cochlear nerve absence on MRI
MRI_Cochl_fibr	MRI	Cat	604 (78%)	Cochlear fibrosis on MRI
MRI_Nerve_Tumor	MRI	Cat	602 (78%)	Nerve tumor on MRI
MRI_Ossif	MRI	Cat	601 (78%)	Ossification on MRI
MRI_done	MRI	Cat	759 (98%)	MRI performed
Sur_ECAP	Surgical	Cat	434 (56%)	ECAP presence at surgery
Sur_Gusher	Surgical	Cat	618 (80%)	Gusher during surgery
Sur_Impedances	Surgical	Cat	423 (55%)	Impedances measured
Sur_Stapedius	Surgical	Cat	402 (52%)	Stapedius reflex
Sur_Status	Surgical	Cat	467 (61%)	Surgical status
Sur_complications_txt	Surgical	Cat	16 (2%)	

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Variable	Category	Type	N (%)	Description
Sur_duration	Surgical	Num	677 (88%)	Surgery duration (min)
Sur_elec_fixation	Surgical	Cat	539 (70%)	Electrode fixation
Sur_elec_insertion	Surgical	Cat	752 (98%)	Electrode insertion
Sur_fac_monitor	Surgical	Cat	672 (87%)	Facial nerve monitoring
Sur_fixation	Surgical	Cat	700 (91%)	Fixation method
Sur_hemorrhage	Surgical	Cat	608 (79%)	Hemorrhage during surgery
Sur_incision_type	Surgical	Cat	701 (91%)	Incision type
Sur_insertion_techn	Surgical	Cat	751 (97%)	Insertion technique
Sur_intracochl_access	Surgical	Cat	754 (98%)	Intracochlear access
Sur_num_of_not_ins_elec	Surgical	Num	210 (27%)	Number of electrodes not inserted
Sur_num_of_short_open_sur	Surgical	Num	94 (12%)	
Sur_periop_medic	Surgical	Cat	692 (90%)	Perioperative medication
Sur_roundwin_findings	Surgical	Cat	696 (90%)	Round window findings
Sur_surg_access	Surgical	Cat	755 (98%)	Surgical access
Postop_Complic_other	Postoperative	Cat	452 (59%)	
Postop_Complic_txt	Postoperative	Cat	6 (1%)	
Postop_Dysgeusia	Postoperative	Cat	456 (59%)	Postop taste disturbance

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Variable	Category	Type	N (%)	Description
Postop_Equili	Postoperative	Cat	456 (59%)	Postop equilibrium issues
Postop_Facialis	Postoperative	Cat	456 (59%)	Postop facial nerve issues
Postop_Hemorrhage	Postoperative	Cat	456 (59%)	Postop hemorrhage
Postop_Implant_failure	Postoperative	Cat	456 (59%)	Implant failure
Postop_Infection	Postoperative	Cat	456 (59%)	Postop infection
Postop_Tinnitus	Postoperative	Cat	456 (59%)	Postop tinnitus
Postop_image	Postoperative	Cat	354 (46%)	Postoperative imaging
Postop_image_result_normal	Postoperative	Cat	331 (43%)	Postop imaging normal
Postop_surg_complications	Postoperative	Cat	456 (59%)	Postop surgical complications
BetreuKlinik	Other	Cat	771 (100%)	Care clinic
C12_poDat	Other	Date	500 (65%)	
CI_Beschreibung	Other	Cat	771 (100%)	CI description
CI_explantiert	Other	Cat	771 (100%)	CI explanted
Chirurg	Other	Num	683 (89%)	Surgeon
Date_Com	Other	Date	771 (100%)	
Date_Eval	Other	Date	771 (100%)	

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Variable	Category	Type	N (%)	Description
Date_SES	Other	Date	771 (100%)	Date of first CI
Datum_ErstesCI	Other	Date	771 (100%)	
Datum_KontralatCI	Other	Date	251 (33%)	
Difference_PTA_OP_monatCI	Other	Cat	704 (91%)	Hospital stay duration
Difference_PTA_OP_monatCo	Other	Num	705 (91%)	
FM_poDat	Other	Date	495 (64%)	
Hosp_Duration	Other	Num	675 (88%)	Implant clinic
ImplKlinik	Other	Cat	771 (100%)	
Interv_Com_OP_mo	Other	Num	771 (100%)	
Interv_Eval_OP_mo	Other	Num	771 (100%)	
Interv_SES_OP_mo	Other	Num	771 (100%)	
KontralatCI_Beschreibung	Other	Cat	251 (33%)	
KontralatCI_ImplKlinik	Other	Cat	251 (33%)	
KontralatCI_Seite	Other	Cat	251 (33%)	
KontralatCI_Typ	Other	Num	251 (33%)	
KontralatCI_explantiert	Other	Cat	251 (33%)	CI model
Modell_CI	Other	Num	771 (100%)	

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Variable	Category	Type	N (%)	Description
Mono1_poDat	Other	Date	464 (60%)	Side of implant (L/R)
Mono2_poDat	Other	Date	464 (60%)	
Mono_bi1_poDat	Other	Date	17 (2%)	
Num_poDat	Other	Date	488 (63%)	
Seite	Other	Cat	771 (100%)	
Todesdatum	Other	Date	31 (4%)	
V08_poDat	Other	Date	510 (66%)	